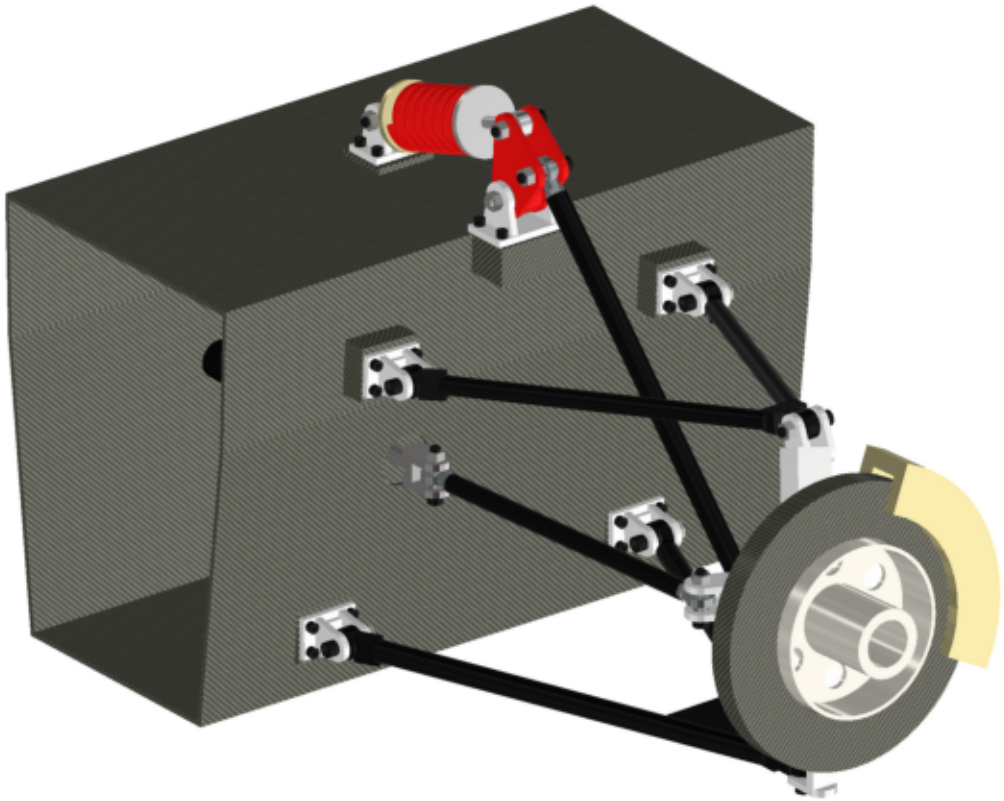
F1Suspension - calculation report

F1-style double-wishbone pushrod front corner. Jay Patel, Stevens Institute of Technology (ME).

Designed and simulated only: every number is calculated or simulated from the CAD model; nothing has been built or tested.



Corner: track 1600 mm, wheelbase 3600 mm, 800 kg car, 305/720 R18 tyre, static camber -3 deg

Geometry: KPI 8.5 deg, caster 6.9 deg, scrub radius 39 mm, trail 31 mm

Camber gain -0.54 deg / 25 mm bump; roll camber (outer) +0.71 deg/deg

Roll centre 72 mm; anti-dive 28 %; bump steer 0.3 mdeg over travel

Installation ratio 0.894 (rising 0.82 -> 1.01); spring 160 N/mm; wheel rate 133 N/mm; ride 4.09 Hz

Mass (Inventor): corner 42.1 kg, unsprung 33.6 kg

Assembly: 126 occurrences, 324 constraints, solver vs Python max error 3e-05 mm, 0 interference over travel and steering

FEA Lower wishbone (4130): worst C2 braking 4.0 g: 412 MPa peak / 266 MPa p99.9, SF 1.12 / 1.73

FEA Upper wishbone (4130): worst C2 braking 4.0 g: 200 MPa peak / 122 MPa p99.9, SF 2.30 / 3.78

FEA Upright (7075-T6): worst C2 braking 4.0 g: 732 MPa peak / 283 MPa p99.9, SF 0.69 / 1.78

FEA Chassis clevis bracket (7075-T6): worst C2 LF: 436 MPa peak / 317 MPa p99.9, SF 1.15 / 1.59

FEA Rocker (7075-T6): worst C4 bump 3 g: 184 MPa peak / 171 MPa p99.9, SF 2.73 / 2.95

1 Design and hardpoints

Brief: an F1-inspired double-wishbone pushrod front suspension corner, modelled in Autodesk Inventor with real joints and geometry, analysed for camber gain, travel, motion ratios and forces, with FEA of the arms and brackets under cornering and braking, plus drawings and a BOM.

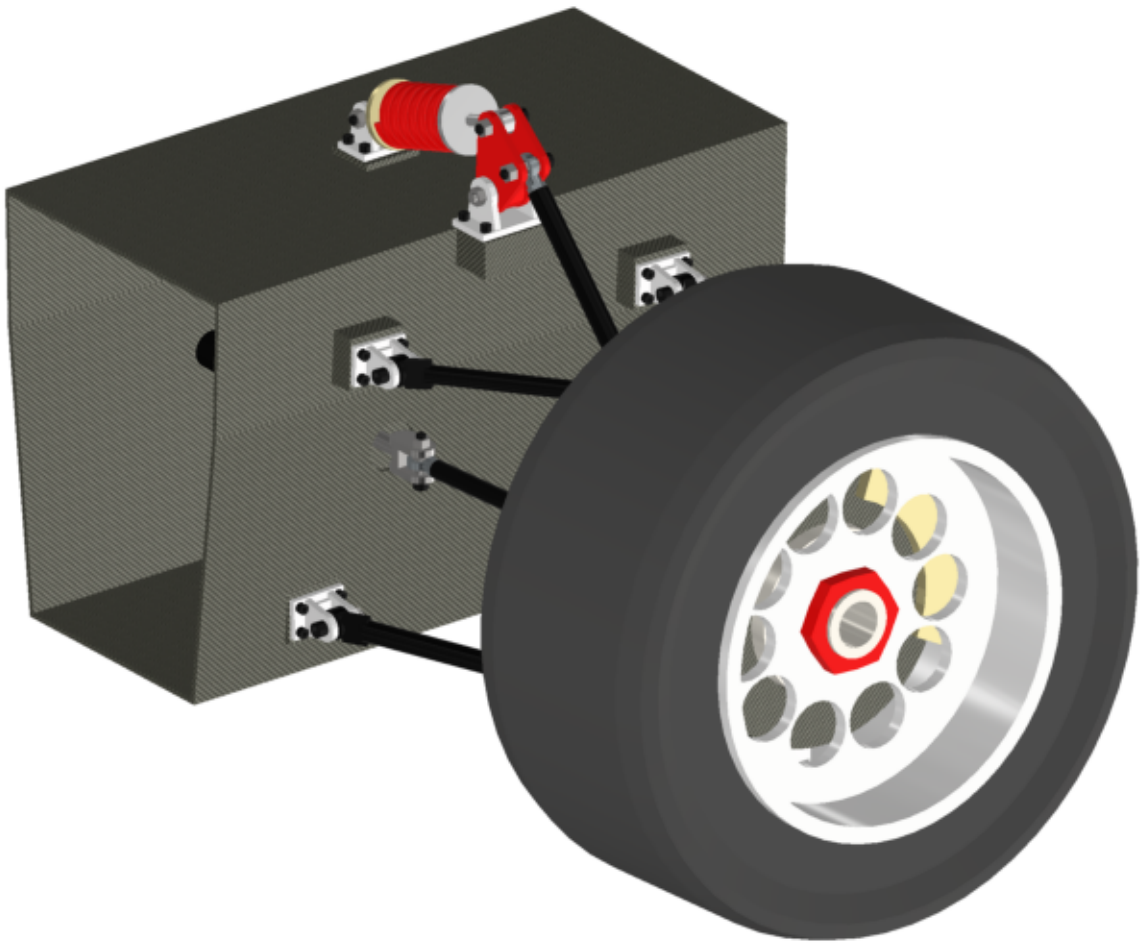
Concept: the corner is modelled at FULL SIZE in car coordinates (ISO 8855: X forward, Y left, Z up; origin on the ground under the front axle). Every part is built from one hardpoint table (_build/design.py), so CAD, kinematics, loads and FEA share the same numbers. Wishbones are 4130 tube with GE 12 E spherical bearings in welded housings; the upright, rocker and brackets are 7075-T6; the pushrod mounts to the upright above the lower ball joint and drives a twin-plate rocker on the chassis top, which compresses a coil-over damper lying across the monocoque. The tie rod runs from the upright steering arm to a rack whose inner point was optimised for zero bump steer.

Design iterations driven by the analysis (documented in Tests/Summary.md): (1) pushrod moved from the lower wishbone to the upright after FEA showed the bracket offset bent the lower legs; (2) lower ball-joint bolt turned vertical so the 13 kN braking load at the LBJ is radial on the spherical bearing instead of axial; (3) leg tubes routed exactly on the pin-to-pin force lines; (4) clevis gaps opened to 24 mm with high-misalignment cone spacers after Inventor interference sweeps found rod-end and housing contact at full steering lock; (5) caliper lugs and the upright's lower arm upsized.

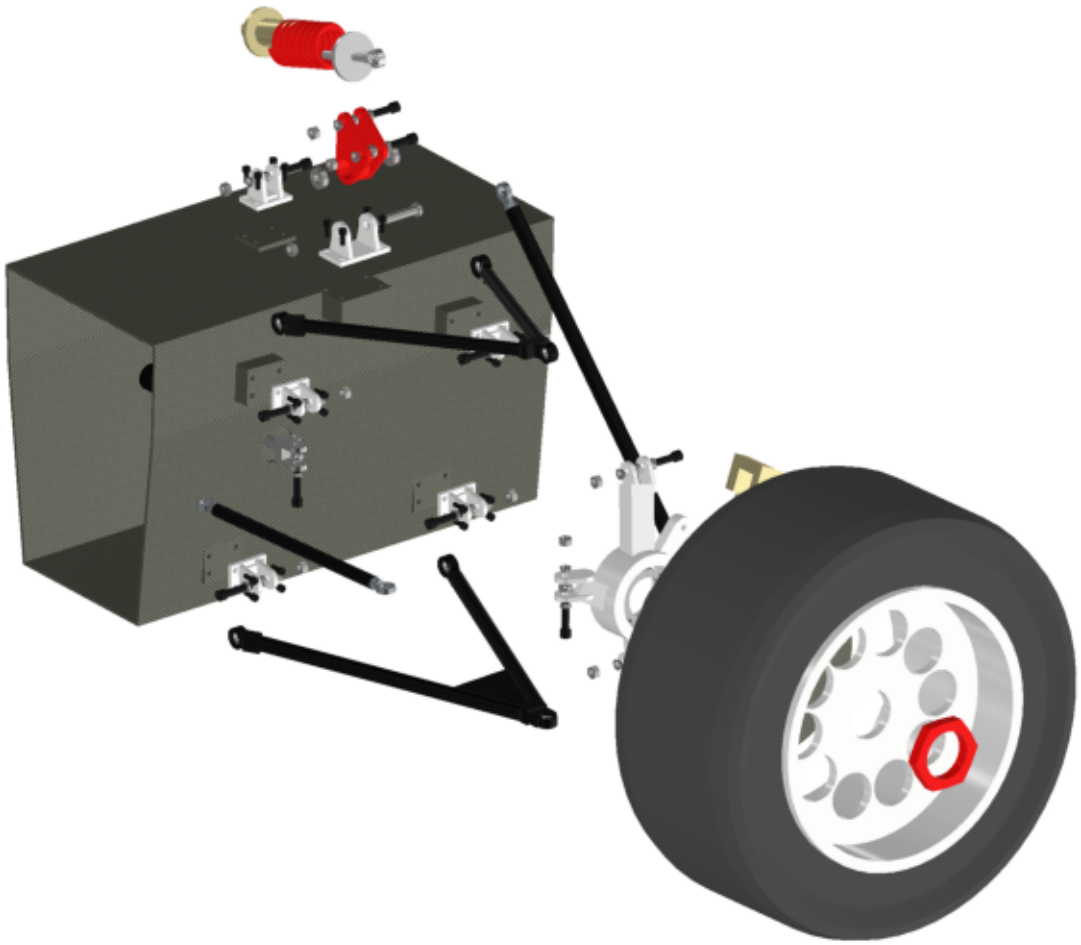
Hardpoints (mm):

point	x	y	z
UF	180.0	265.0	506.0
UR	-220.0	265.0	494.0
UBJ	-35.0	680.0	545.0
LF	220.0	205.0	186.0
LR	-200.0	205.0	194.0
LBJ	10.0	735.0	175.0
TO	125.0	703.0	392.0
TI	125.0	233.5	367.7
PO	-5.0	712.0	232.0
RP	-5.0	215.0	660.0
DC	-5.0	-18.0	668.0
RPP	-5.0	258.2	701.7
RDP	-5.0	212.2	740.0

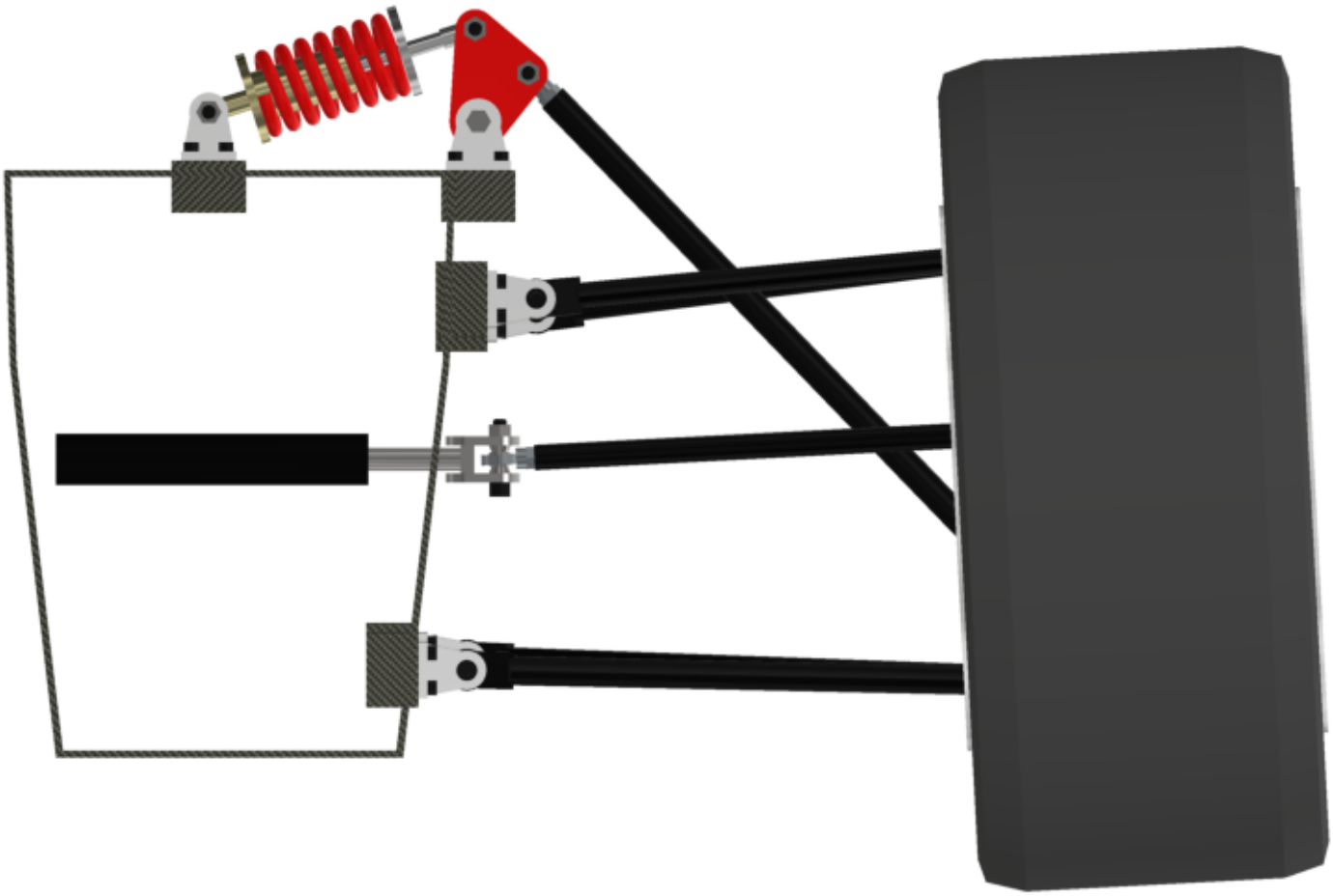
Isometric, static ride height (Inventor, Realistic style)



Exploded view (F1Suspension_Exploded.iam)



Front view



2 Kinematics, motion ratio, rates

Method: rigid-body kinematics (design.Kin) solves the upright pose from the five link-length closures (two legs of each wishbone plus the tie rod) and the wheel-centre height, by least squares in 4 mm steps; the pushrod, rocker and damper follow. Roll centre and anti-dive come from the velocity of the contact-patch point fixed to the upright (front- and side-view instant centres). The roll sweep combines left (bump) and mirrored right (rebound) wheels.

Results: static camber -3.0 deg; -3.76 deg at +35 mm and -2.50 deg at -25 mm (gain -0.0206 deg/mm). Half-track change +2.2 / -2.8 mm. Roll centre 72 mm static, 62 mm at full bump, 80 mm at full rebound; it moves 201 mm sideways per degree of roll. Anti-dive 28 % (brake bias 57% front, CG 300 mm).

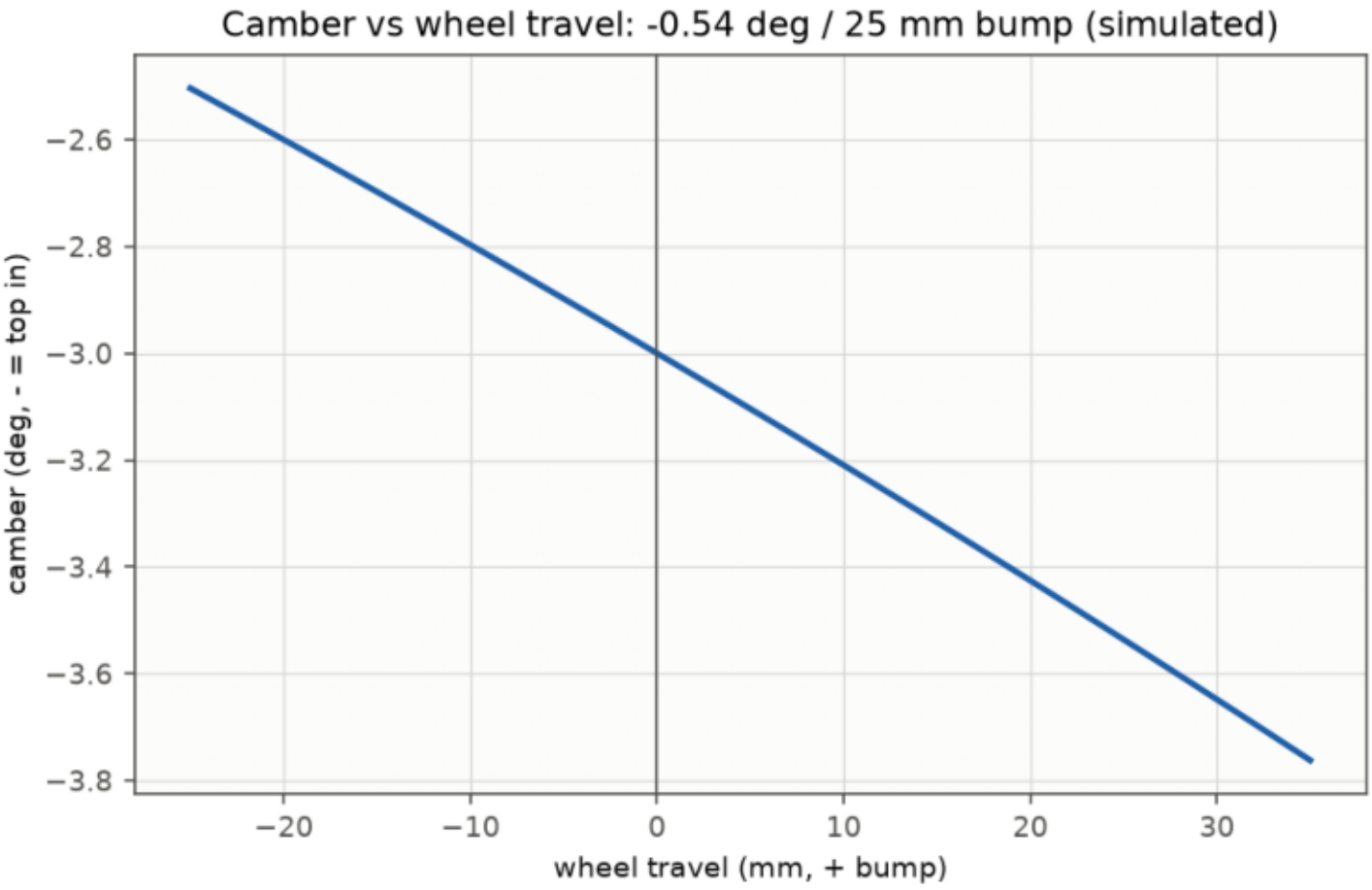
Bump steer: the inner tie-rod point was optimised (Nelder-Mead on y, z) to 0.26 mdeg toe change over the full travel with the rack centred. Steering: +-28 mm rack gives 11.5 deg road-wheel angle.

Motion ratios: pushrod ratio 0.696 (rocker input arc per mm wheel), rocker ratio 1.286 (damper per rocker input), installation ratio 0.894 static, rising from 0.816 to 1.009. The damper uses 54.7 of its 60 mm stroke.

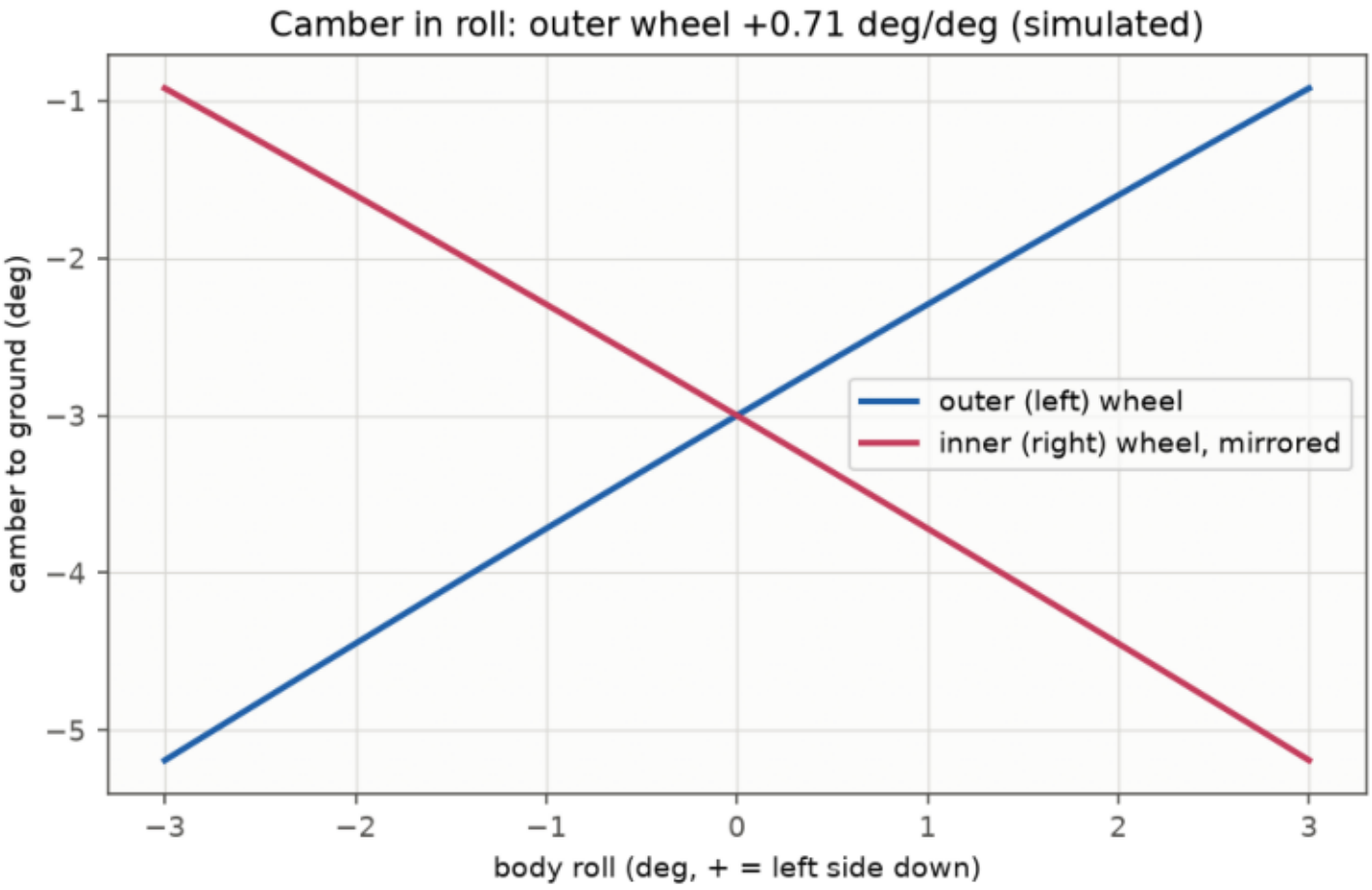
Springing: target front ride frequency 4 Hz (sprung corner 146 kg, tyre 350 N/mm). Spring d 12 mm wire, Dm 64 mm, 4.9 active coils -> 160 N/mm (914 lbf/in); wheel rate k_s IR^2 + F_s dIR/dz = 133 N/mm static, 174 N/mm at full bump; ride 4.09 Hz. Preload 1606 N (10.0 mm), free length 139 mm, shortest 96 mm vs solid 83 mm; shear stress at full bump 844 MPa (Wahl), SF 1.24 vs 0.55 Su (assumed Si-Cr). A real F1 car adds heave springs / torsion bars and an anti-roll bar; those are not modelled.

Verification: the Inventor assembly is driven by the Heave and Steer parameters; at 19 poses the solved occurrence poses match this model to 1e-05 deg camber and 3e-05 mm at the hardpoints (two independent solvers).

Camber vs wheel travel



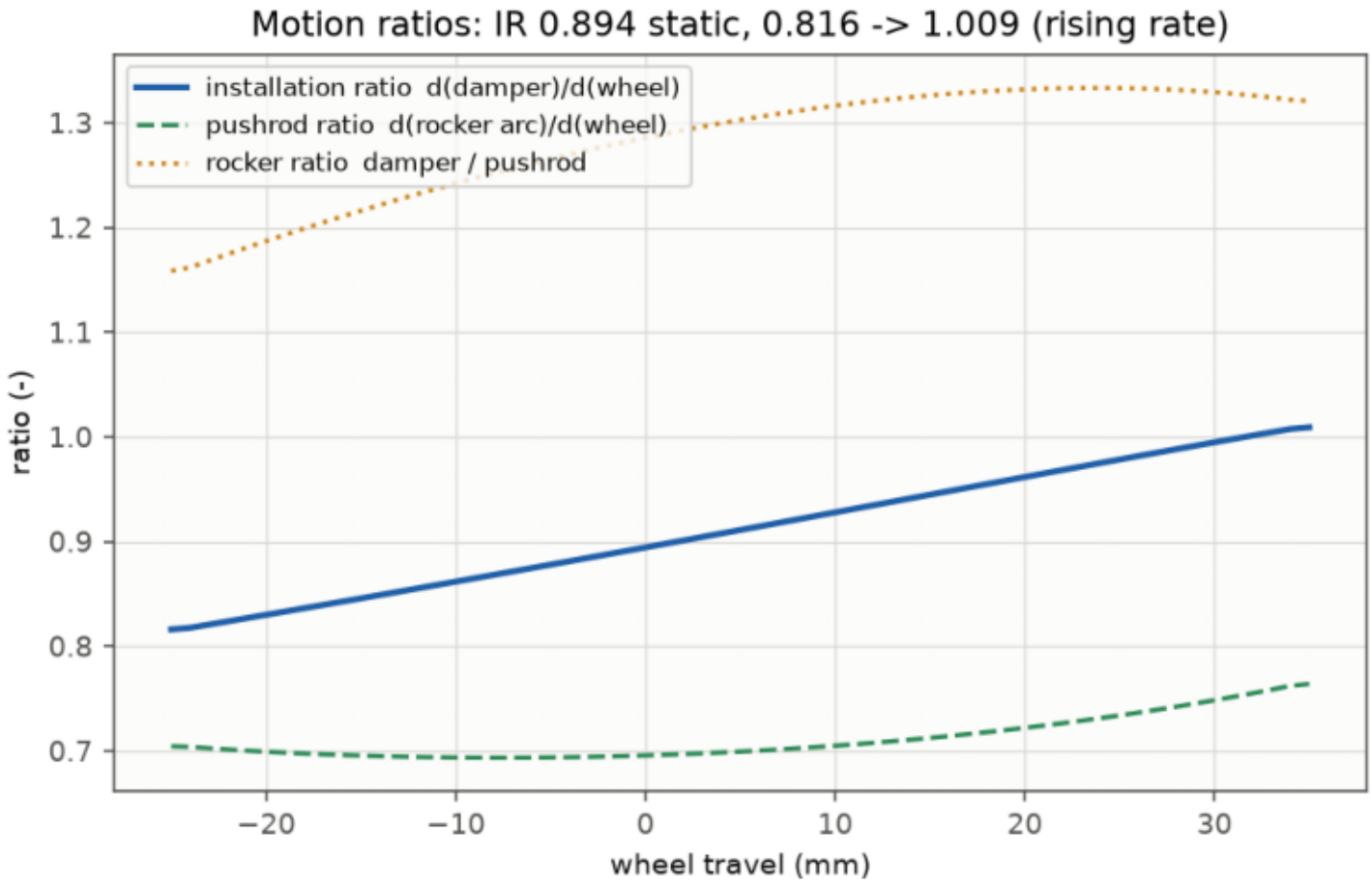
Camber in roll



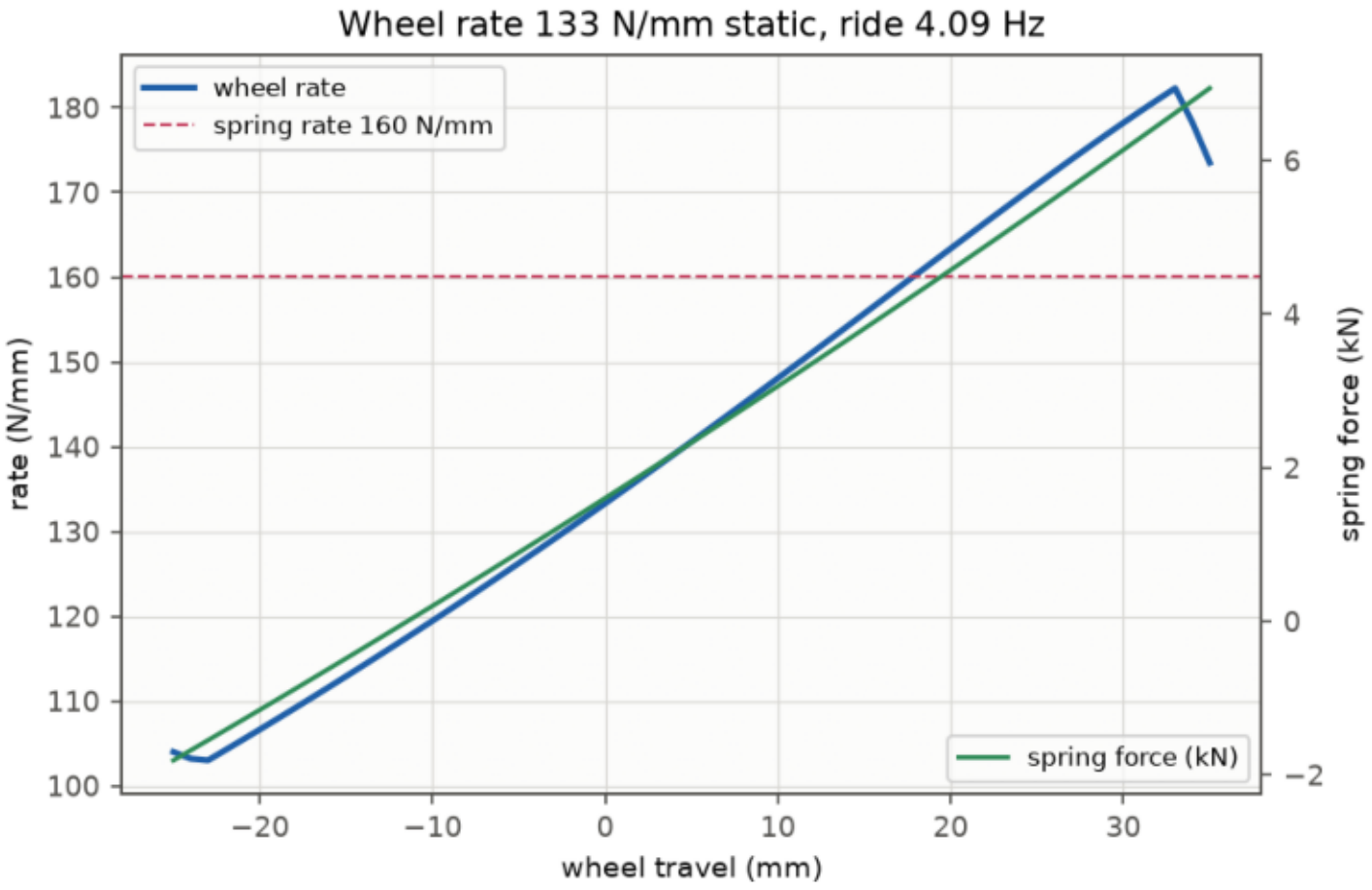
Bump steer


Bump steer: 0.26 mdeg over full travel (tie-rod inner point optimised)
toe (deg, + = toe-in)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−20
−10
0
10
20
30
wheel travel (mm)

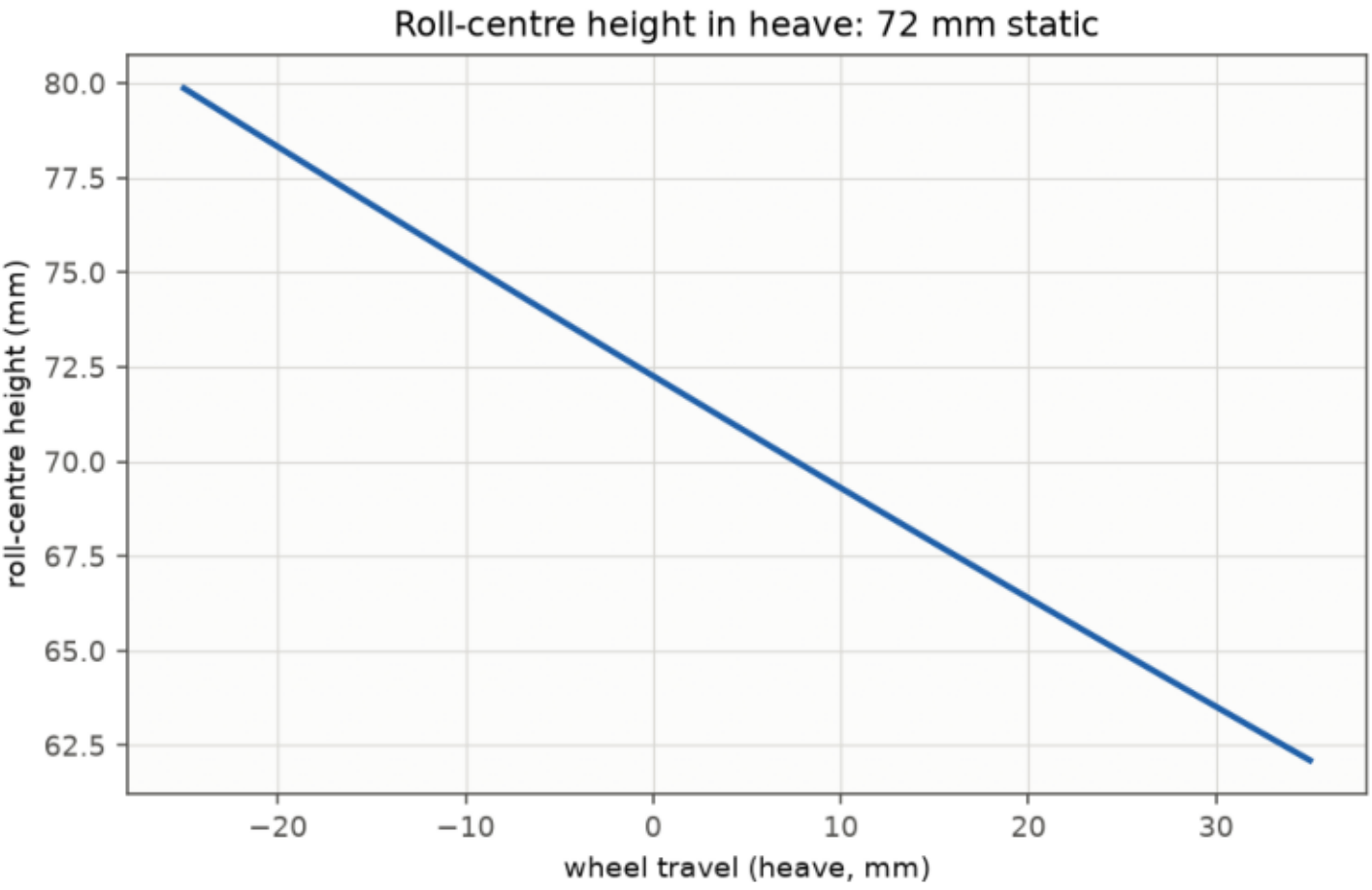
Motion ratios



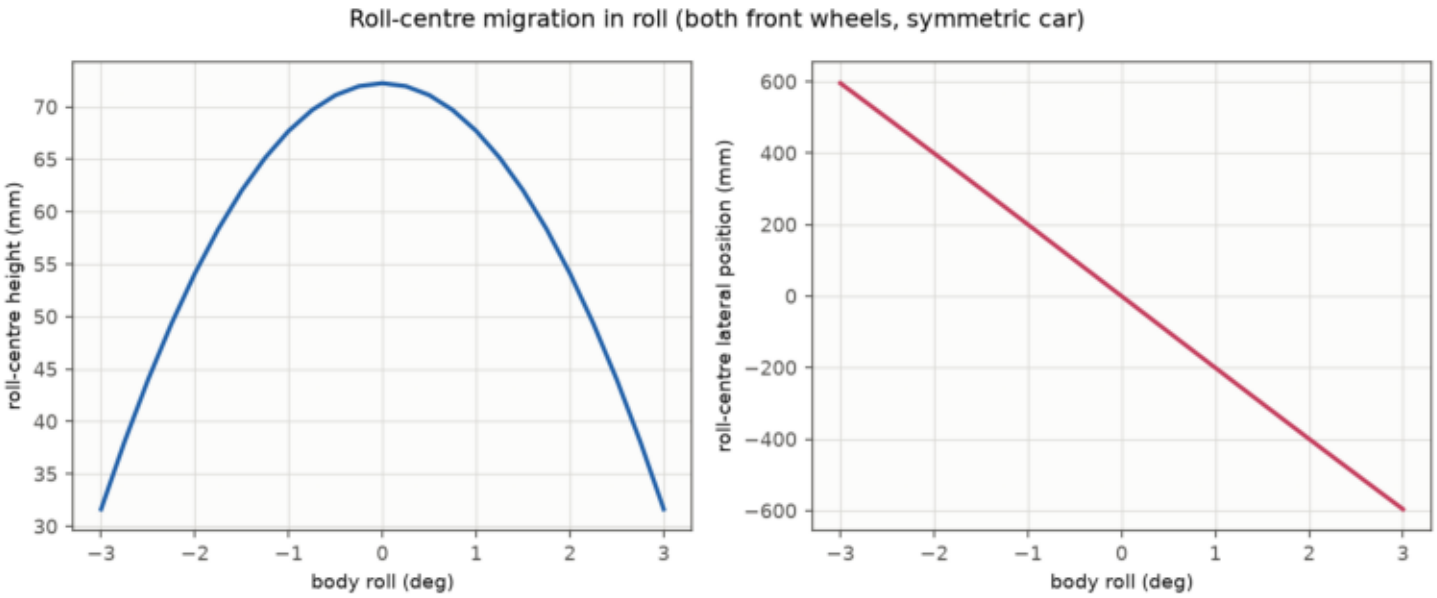
Wheel rate and spring force



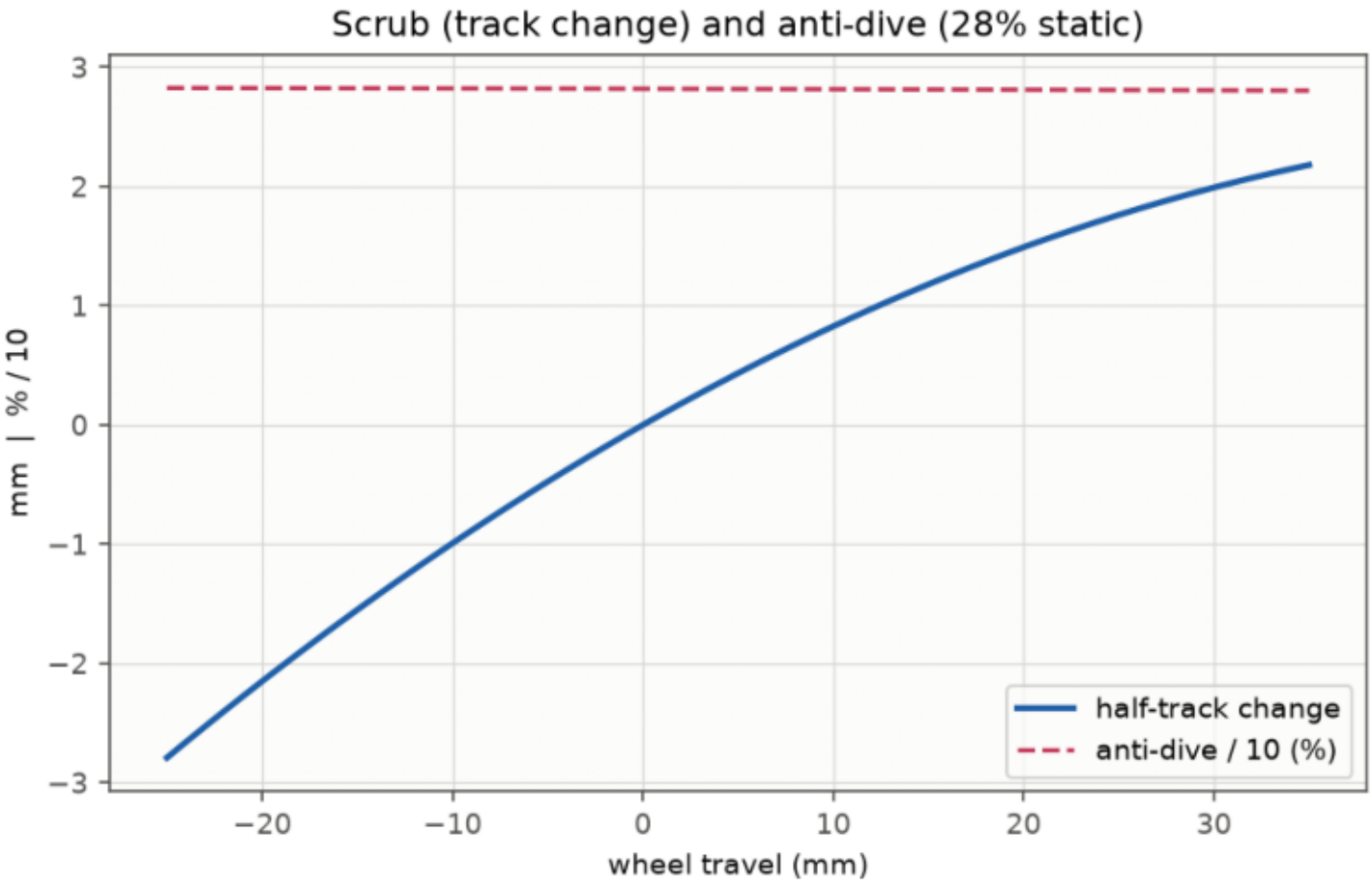
Roll-centre height in heave



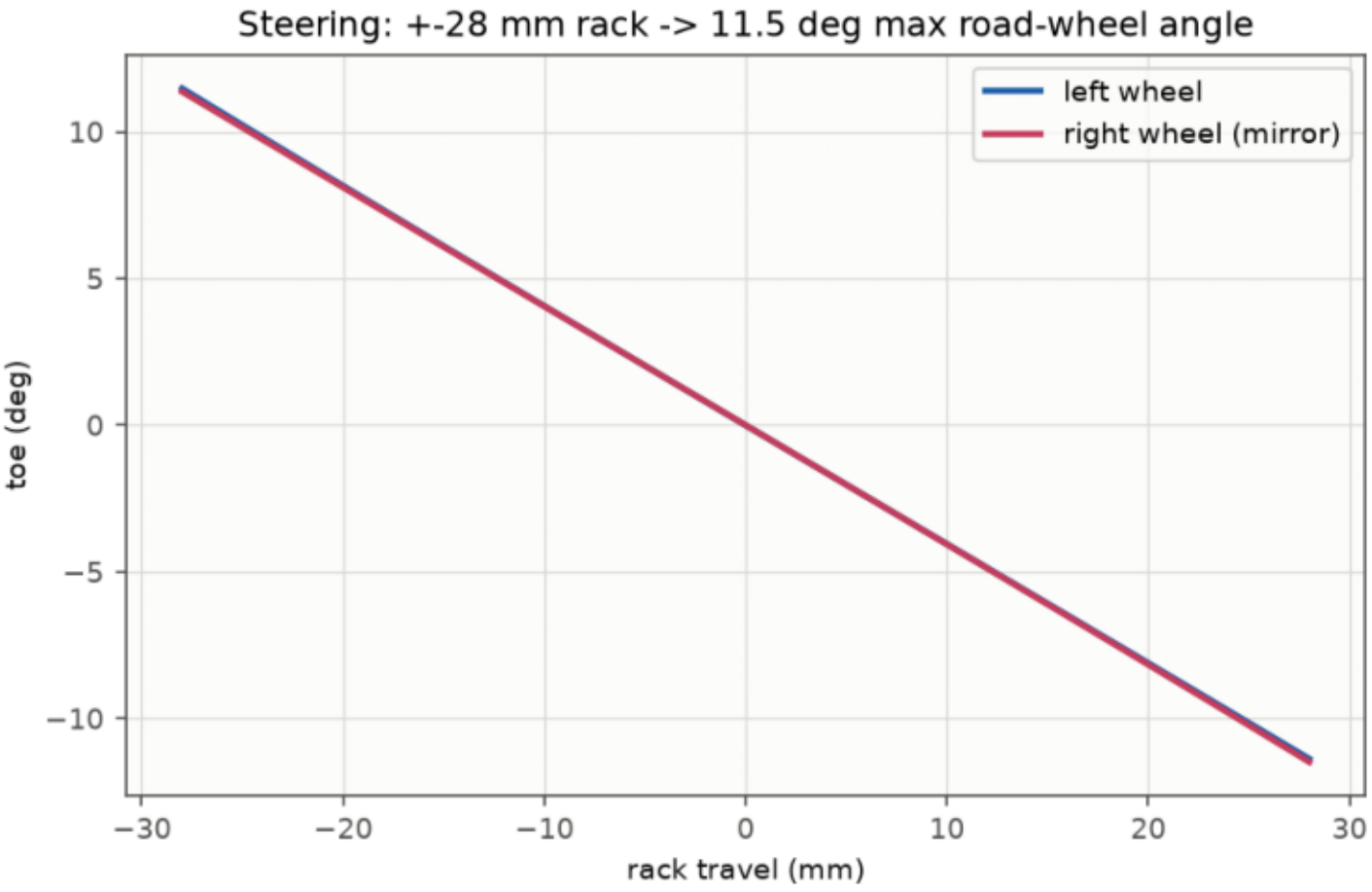
Roll-centre migration in roll



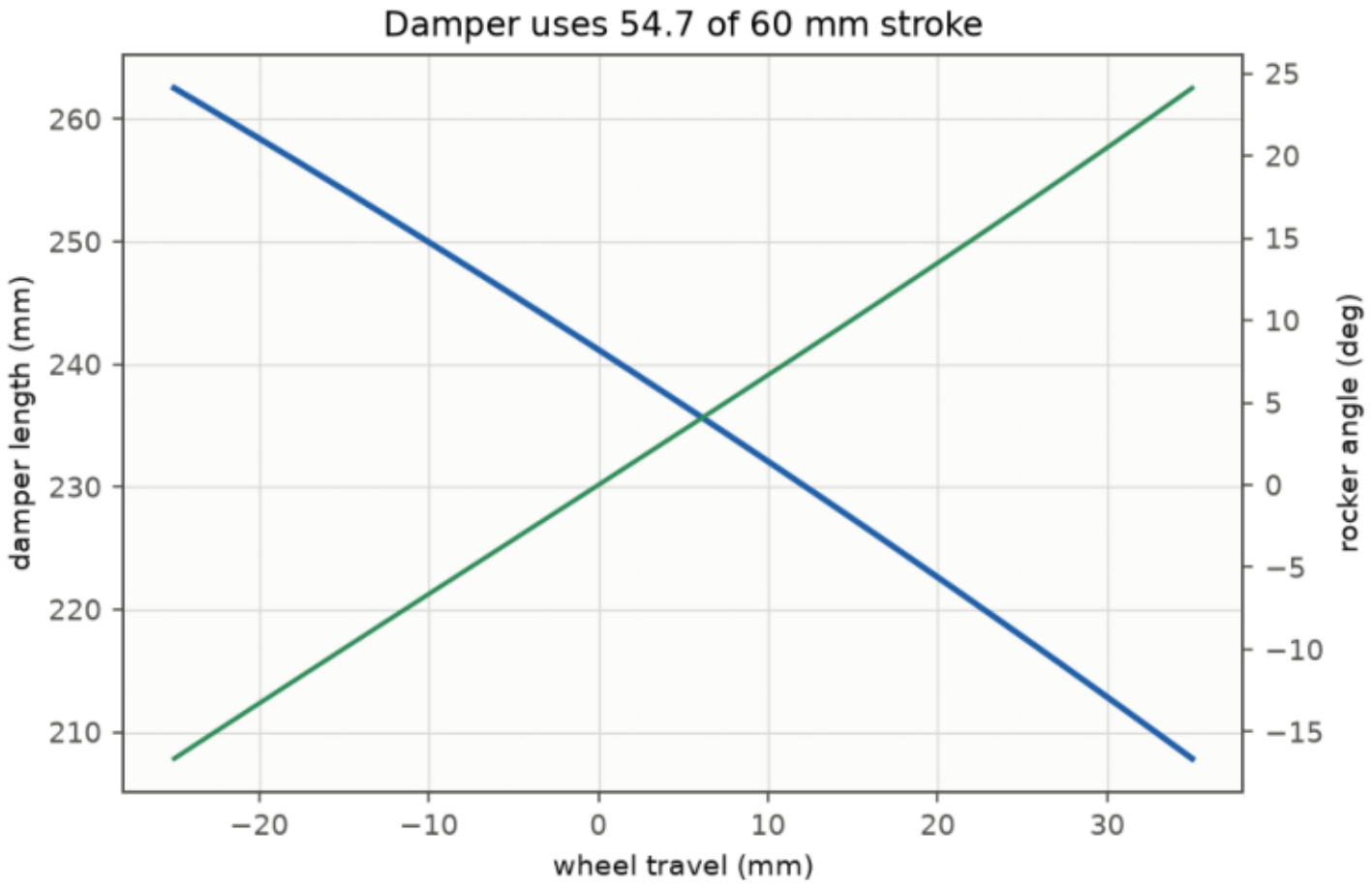
Track change and anti-dive



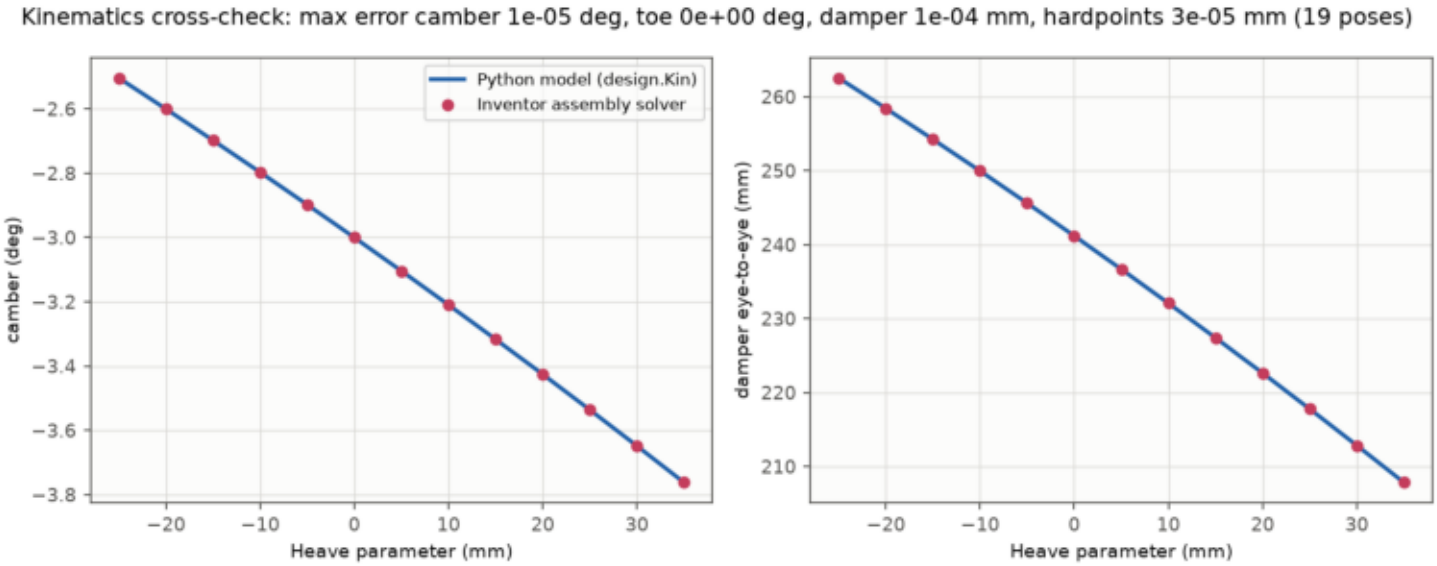
Steering


Steering: +-28 mm rack -> 11.5 deg max road-wheel angle
left wheel
right wheel (mirror)
toe (deg)
10
5
0
−5
−10
−30
−20
−10
0
10
20
30
rack travel (mm)

Damper and rocker travel



Inventor solver vs Python model



3 Loads, member forces, buckling

Load cases (quasi-static limit loads on the outer front tyre, point-mass vehicle with aero): C1 cornering 4.0 g: 4.0 g lateral, downforce 2.0 W (40% front), 55% front roll-stiffness share; C2 braking 4.0 g: 4.0 g deceleration, downforce 2.0 W, brake bias 57% front, outboard brakes; C3 combined 2.5 g + 2.5 g: trail braking: 2.5 g lateral + 2.5 g braking, downforce 1.5 W; C4 bump 3 g: kerb strike: 3 x (static + high-speed aero) vertical; S0 static: car at rest, no aero.

Member forces: the upright is held by six two-force members (two legs per wishbone, tie rod, pushrod), so one 6x6 equilibrium gives every axial force exactly (residual < 1e-11 N). Forces in kN, + = tension:

case	Fx	Fy	Fz	UF	UR	LF	LR	tie	push	damper
C1 cornering 4.0 g	0.0	-10.9	8.1	1.9	1.9	-4.7	-4.4	1.4	-10.4	8.1
C2 braking 4.0 g	-8.9	0.0	6.2	-5.7	3.9	20.9	-14.7	1.7	-8.5	6.6
C3 combined 2.5 g + 2.	-5.6	-7.0	7.0	-2.4	3.7	8.8	-13.4	2.2	-8.8	6.9
C4 bump 3 g	0.0	0.0	14.7	0.2	-0.4	8.7	9.4	-1.9	-21.3	16.6
S0 static	0.0	0.0	1.8	0.0	-0.0	1.0	1.1	-0.2	-2.6	2.0

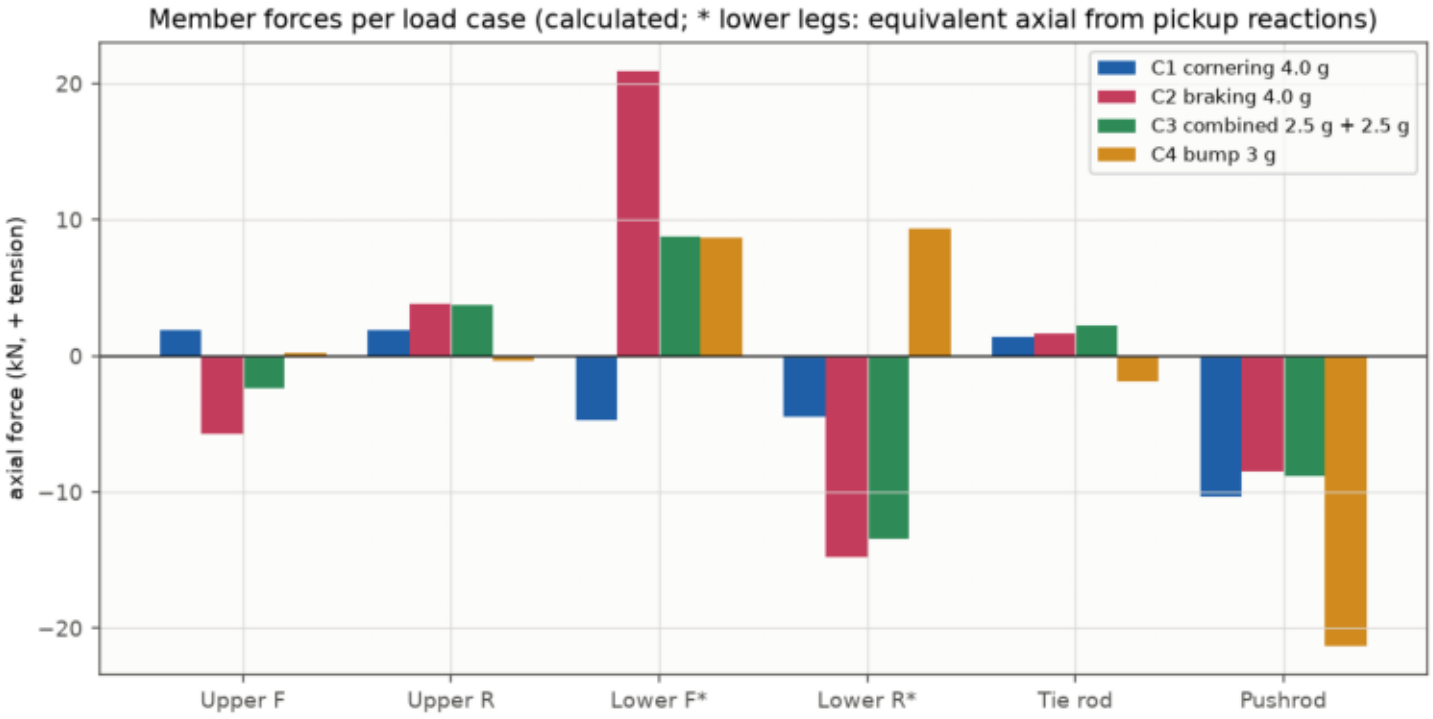
Buckling (pinned-pinned, Euler or Johnson by slenderness) and axial stress, 4130 normalised Sy 460 MPa:

member	tube	L	max T	max C	Pcr	SF buck	SF yield
Upper front leg	22.2 x 1.65	469	1.9	-5.7	37.5	6.55	8.57
Upper rear leg	22.2 x 1.65	457	3.9	-0.4	38.0	101.58	12.68
Lower front leg	25.4 x 1.65	570	20.9	-4.7	41.9	8.85	2.71
Lower rear leg	25.4 x 1.65	570	9.4	-14.7	41.8	2.84	3.84
Tie rod	22.2 x 1.65	470	2.2	-1.9	37.4	19.99	21.98
Pushrod	25.4 x 2.11	653	-2.6	-21.3	45.8	2.15	3.33

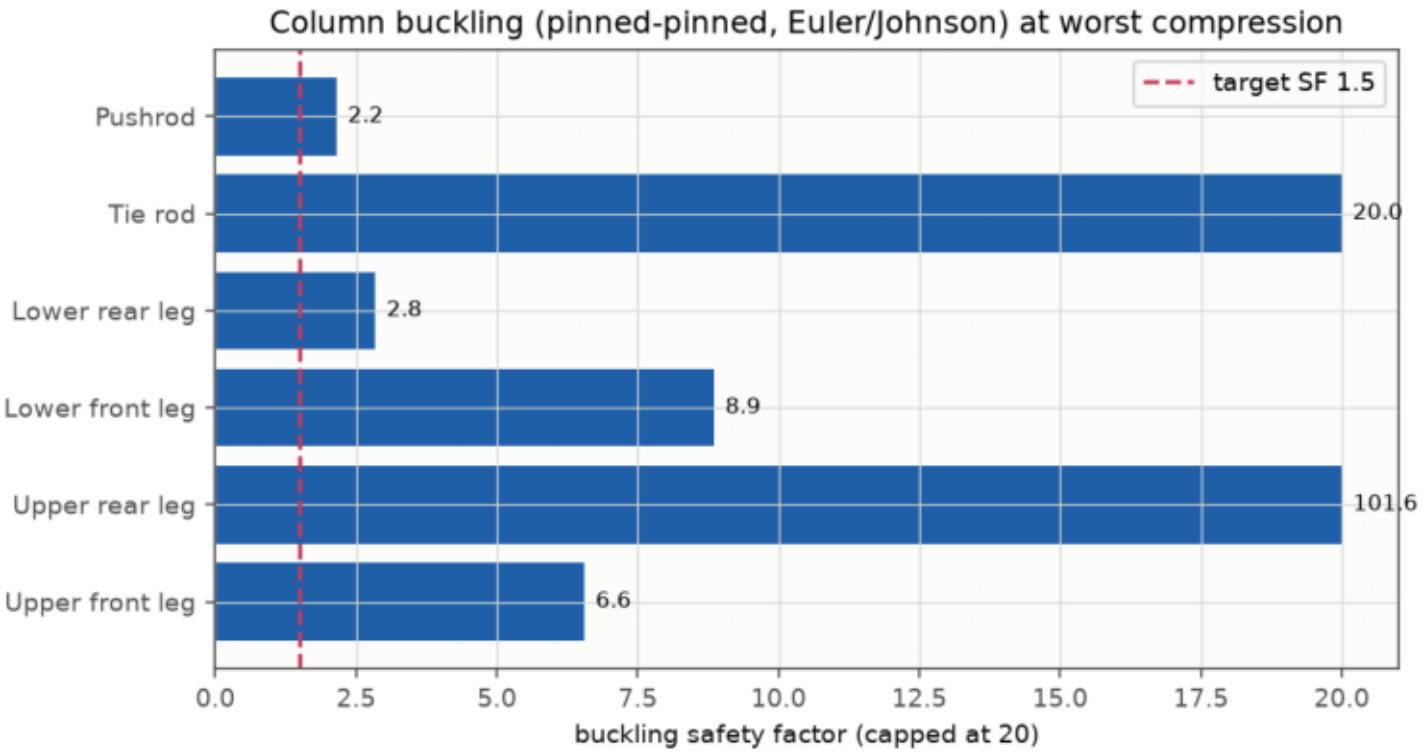
Joint hardware: largest joint force 21.3 kN; M12 12.9 bolt in double shear SF 6.9; GE 12 E static SF 2.54; rod end SF 1.64. C0 of GE 12 E from the SKF catalogue class (~54 kN); rod-end static radial capacity 35 kN ASSUMED (SI 12 class) - check the chosen supplier's datasheet.

Braking with outboard brakes: the wheel is treated as locked to the upright; the brake torque is reacted by the caliper on the upright lugs (pad force at r 146 mm), the rest of the tyre force goes through the hub bearing.

Member forces per load case



Buckling safety factors



4 Structural FEA (simulated)

Solver: gmsh (OpenCASCADE STEP import of the Inventor parts) + scikit-fem linear elasticity, quadratic P2 tetrahedra, PyPardiso direct solve with all load cases as one multi-RHS solve (same chain as QuadDrone / RoboticArm6DOF).

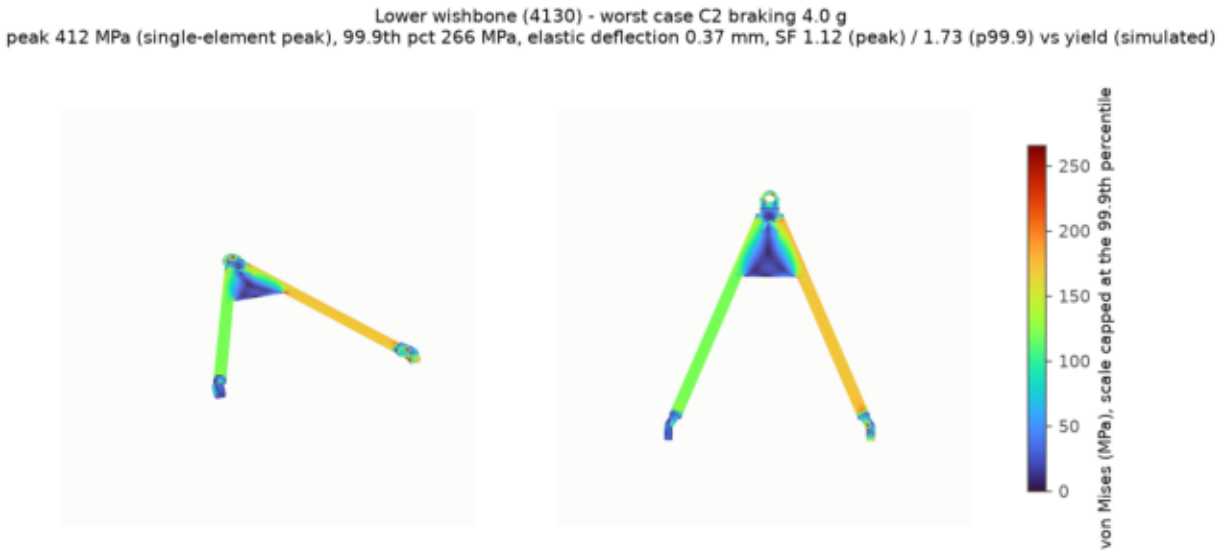
Loads: the joint forces of each load case are applied as cosine bearing pressure on the half of each bore the pin pushes on, plus uniform axial shear. Each free part's force set is self-equilibrated and rigid-body motion is removed with a statically determinate 3-2-1 point support; the support reactions (reported) are < 1 % of the loads, so the spherical joints are not artificially clamped. The upright gets the hub-bearing load as two rows (force + moment) and the caliper reaction split over its two lugs so both force and moment about the wheel centre are reproduced. The clevis bracket is bonded at its base and every pickup reaction of every case is solved as its own column.

Peak vs design value: the CAD has sharp re-entrant edges where real parts have weld or machining fillets, so the single highest element is a mesh singularity. Both the peak and the 99.9th-percentile element stress are reported; the safety factor quoted for design is on the 99.9th percentile, and the peak SF is shown next to it. Target SF >= 1.5 on yield at limit load.

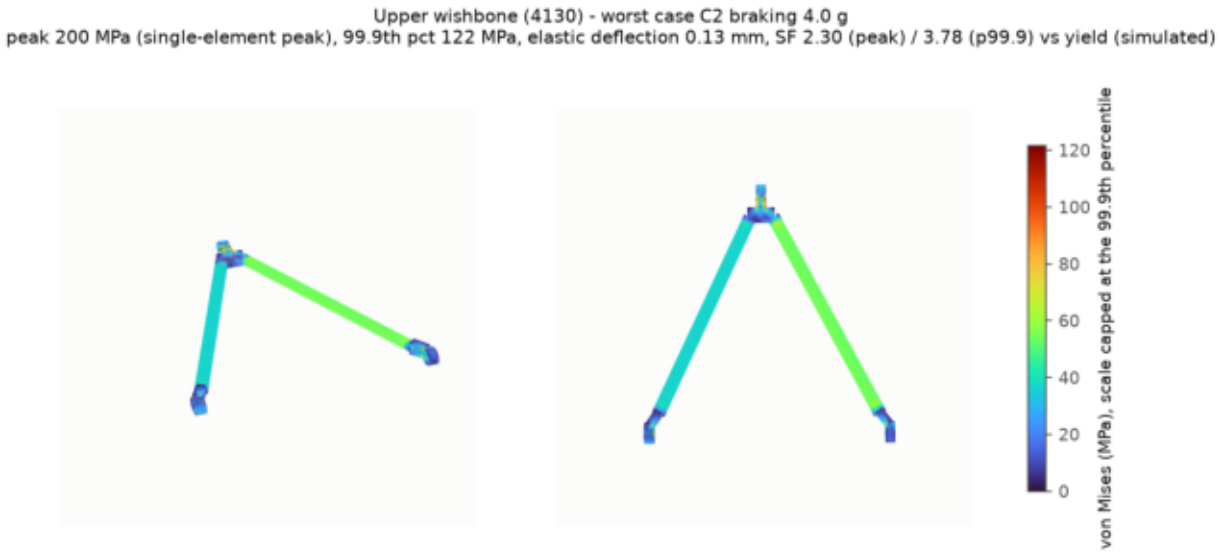
part	case	DOF	peak	p99.9	u el.	SF peak	SF p99.9
LowerWishbone	C1 cornering 4.0 g	520,863	117	59	0.11	3.95	7.75
LowerWishbone	C2 braking 4.0 g	520,863	412	266	0.37	1.12	1.73
LowerWishbone	C3 combined 2.5 g + 2.	520,863	223	143	0.22	2.06	3.21
LowerWishbone	C4 bump 3 g	520,863	235	194	0.22	1.95	2.37
UpperWishbone	C1 cornering 4.0 g	329,403	82	66	0.04	5.61	6.94
UpperWishbone	C2 braking 4.0 g	329,403	200	122	0.13	2.30	3.78
UpperWishbone	C3 combined 2.5 g + 2.	329,403	147	79	0.08	3.12	5.80
UpperWishbone	C4 bump 3 g	329,403	12	7	0.01	37.83	62.51
Upright	C1 cornering 4.0 g	482,826	594	286	1.19	0.85	1.76
Upright	C2 braking 4.0 g	482,826	732	283	1.62	0.69	1.78
Upright	C3 combined 2.5 g + 2.	482,826	598	257	1.23	0.84	1.96
Upright	C4 bump 3 g	482,826	472	245	1.30	1.07	2.06
WishboneClevis	C1 UF	246,786	49	36	0.04	10.36	13.93
WishboneClevis	C1 LF	246,786	100	72	0.09	5.03	6.98
WishboneClevis	C1 LR	246,786	89	68	0.08	5.64	7.39
WishboneClevis	C1 DC	246,786	113	97	0.02	4.47	5.21
WishboneClevis	C2 UF	246,786	145	107	0.13	3.47	4.69
WishboneClevis	C2 LF	246,786	436	317	0.37	1.15	1.59
WishboneClevis	C2 LR	246,786	296	226	0.27	1.70	2.23
WishboneClevis	C2 DC	246,786	92	79	0.02	5.48	6.39
WishboneClevis	C3 UF	246,786	60	44	0.05	8.39	11.34
WishboneClevis	C3 LF	246,786	183	133	0.15	2.75	3.78
WishboneClevis	C3 LR	246,786	269	206	0.24	1.87	2.45
WishboneClevis	C3 DC	246,786	96	82	0.02	5.25	6.12
WishboneClevis	C4 UF	246,786	5	4	0.00	92.90	124.91
WishboneClevis	C4 LF	246,786	182	132	0.15	2.77	3.81
WishboneClevis	C4 LR	246,786	187	144	0.17	2.69	3.48
WishboneClevis	C4 DC	246,786	231	198	0.04	2.18	2.54
Rocker	C1 cornering 4.0 g	288,402	90	83	0.04	5.61	6.06
Rocker	C2 braking 4.0 g	288,402	73	68	0.03	6.87	7.43

Rocker	C3 combined 2.5 g + 2.	288,402	76	71	0.04	6.58	7.11
Rocker	C4 bump 3 g	288,402	184	171	0.09	2.73	2.95

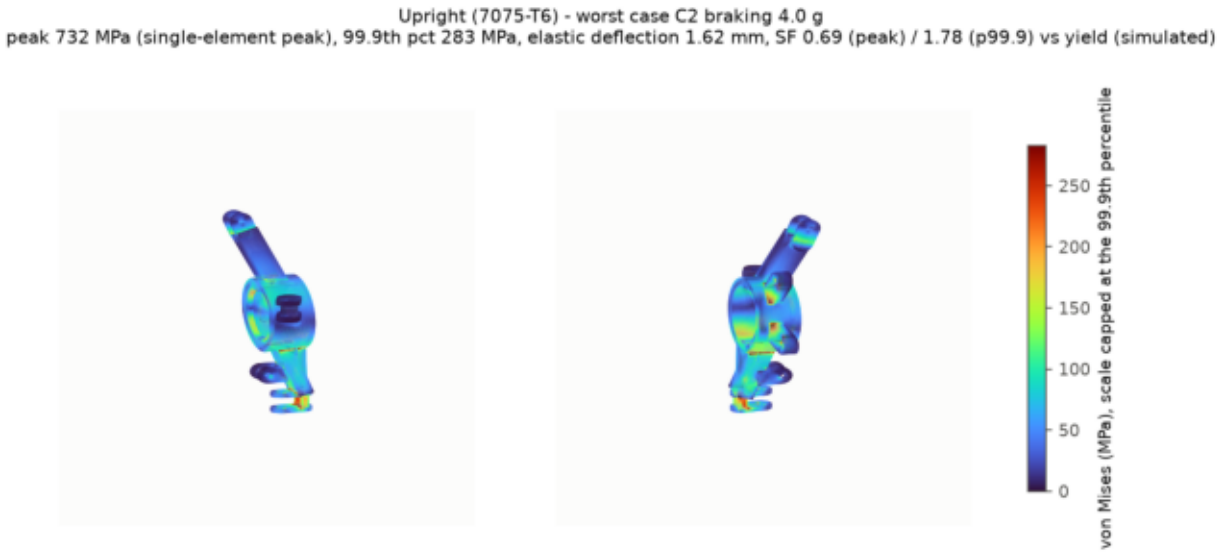
FEA - Lower wishbone (4130)



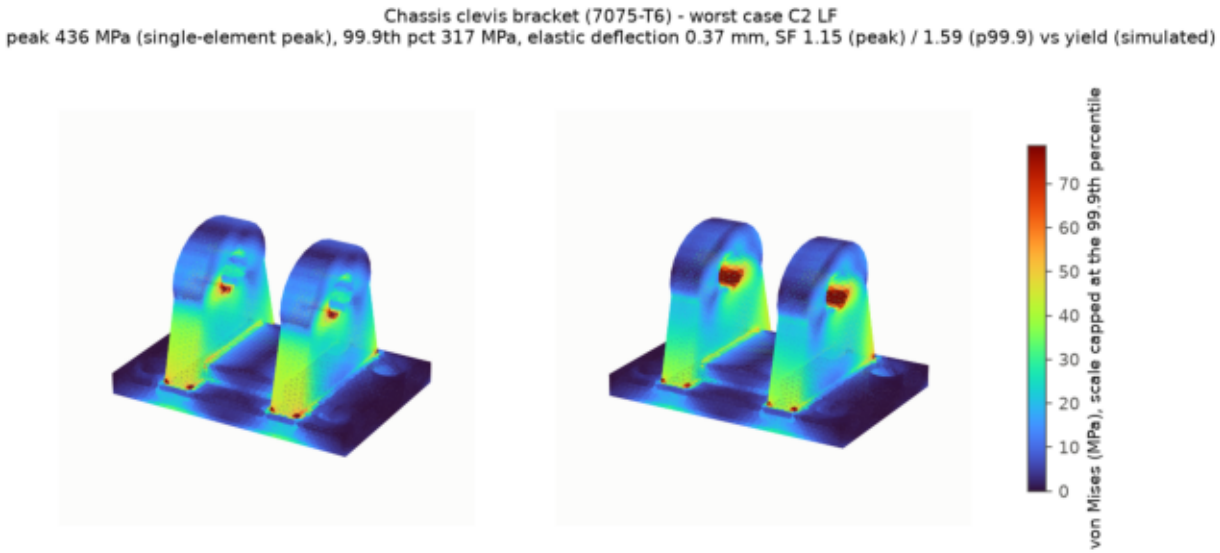
FEA - Upper wishbone (4130)



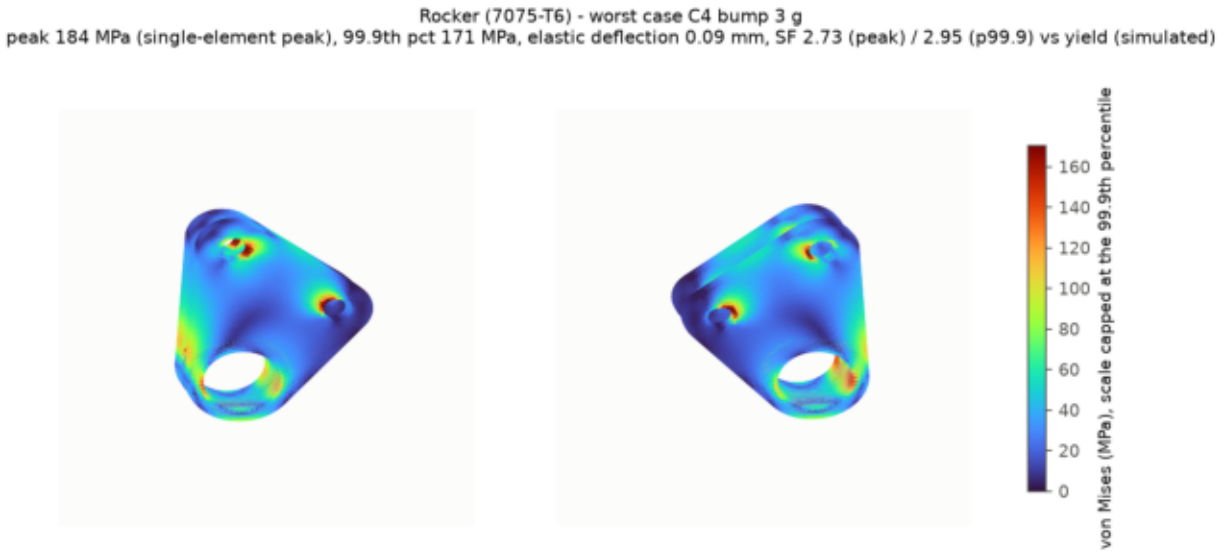
FEA - Upright (7075-T6)



FEA - Chassis clevis bracket (7075-T6)



FEA - Rocker (7075-T6)



5 Assembly checks (Inventor)

Inventor checks: 126 occurrences (33 files), 324 constraints {'Flush': 233, 'Insert': 24, 'Mate': 63, 'Angle': 4}, unhealthy 0, DOF histogram {'0T/0R': 126}.

Interference (AnalyzeInterference on all occurrences, > 0.01 mm3) at 19 poses from full rebound to full bump and at full lock: 0 clashing pairs. The coil spring is modelled at its static installed length, so its overlap with the shaft perch in bump is listed separately and is expected (its compression is checked analytically in section 2).

The checks found and drove fixes for rod-end / clevis contact at full lock, damper perch / rocker contact in rebound, rack clevis / chassis contact at full lock and a rim / upper-wishbone touch at droop plus full lock (rack travel limited to +-28 mm).

6 Mass (Inventor)

Masses are Inventor mass properties at the material densities, with datasheet-class estimates as mass overrides on the envelope parts (tyre 10 kg, damper, hub bearing, disc, caliper).

class	mass kg
half	6.15
sprung	5.39
unsprung	30.53

Corner total 42.08 kg; unsprung 33.61 kg (links counted 50 %).

7 Assumptions and limitations

Designed and simulated only. No part of this corner has been built or tested; every number is a model prediction.

Vehicle: 800 kg, 45 % front, CG 300 mm, downforce 2.0 x weight at the case speed with 40 % front balance, 55 % front roll-stiffness share, brake bias 57 % front. Tyre: rigid 360 mm radius for kinematics, 350 N/mm vertical stiffness (assumed).

Load cases are quasi-static limit loads; no dynamic amplification, fatigue or damping forces are included. Tyre aligning and overturning moments are neglected.

Purchased items without step.parts models (spherical bearings, rod ends, hub bearing, damper, caliper, disc, rim, tyre) are catalogue-dimension or envelope models; their masses are estimates and their ratings must be checked against the supplier datasheet (GE 12 E C0 ~54 kN, rod end 35 kN assumed). ISO 4762 screws, M12 locknuts and 6202 bearings are step.parts STEP files.

FEA: linear elastic, isotropic, nominal geometry, no weld fillets, bonded where parts are welded; weld/HAZ allowable 360 MPa noted for the 4130 wishbones. The spherical bearings are idealised as bearing-pressure loads on the bores.

The chassis section and rack housing are reference geometry. Heave springs, anti-roll bar, uprights' brake ducts and the right-hand corner are not modelled (the right damper would be staggered in X to clear this one).